Chapter 2: Algebraic Numbers

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Exercise 2.1

a) Algebraic, not integer.

b) Algebraic integer.

c) Algebraic integer.

d) Algebraic integer.

e) Algebraic, not integer.

f) Algebraic integer.

Exercise 2.2

See *Example 2.3*, we can easily get $\mathbb{Q}(\sqrt{3}, \sqrt[3]{5}) = \mathbb{Q}(\sqrt{3} + \sqrt[3]{5})$.

Exercise 2.3

Notice the minimal polynomial of $\sqrt[3]{7}$ over $\mathbb{Q}$ is $x^3 - 7$, which have roots

$$\sqrt[3]{7},\ \sqrt[3]{7}\omega,\ \sqrt[3]{7}\omega^2,\ \text{where } \omega \text{ is the 3rd root of unity.}$$

Therefore, the 3 monomorphisms will be

$$\begin{cases} \sigma_1 : \sigma_1(a) = a, \sigma_1(\sqrt[3]{7}) = \sqrt[3]{7} \\ \sigma_2 : \sigma_2(a) = a, \sigma_2(\sqrt[3]{7}) = \sqrt[3]{7}\omega \\ \sigma_3 : \sigma_3(a) = a, \sigma_3(\sqrt[3]{7}) = \sqrt[3]{7}\omega^2 \end{cases}, \text{ where } a \text{ is an arbitary number in } \mathbb{Q}.$$

Exercise 2.4

Let monomorphisms of $\mathbb{Q}(\sqrt{3}, \sqrt{5}) = \mathbb{Q}(\sqrt{3} + \sqrt{5}) \longrightarrow \mathbb{C}$ be $\sigma_k, k \in \{1, 2, 3, 4\}$ which holds its value in $\mathbb{Q}$ and

$$\begin{cases} \sigma_1 : \sigma_1(\sqrt{3} + \sqrt{5}) = \sqrt{3} + \sqrt{5} \\ \sigma_2 : \sigma_2(\sqrt{3} + 2\sqrt{5}) = -\sqrt{3} - \sqrt{5} \\ \sigma_3 : \sigma_3(\sqrt{3} + 2\sqrt{5}) = \sqrt{3} - \sqrt{5} \\ \sigma_4 : \sigma_4(\sqrt{3} + 2\sqrt{5}) = -\sqrt{3} + 2\sqrt{5} \end{cases}$$

Hence, the discriminant of $\mathbb{Q}(\sqrt{3} + \sqrt{5})$ is

$$\det \begin{bmatrix} 1 & \sqrt{3} + \sqrt{5} & 8 + 2\sqrt{15} & 18\sqrt{3} + 14\sqrt{5} \\ 1 & -\sqrt{3} - \sqrt{5} & 8 + 2\sqrt{15} & -18\sqrt{3} - 14\sqrt{5} \\ 1 & \sqrt{3} - \sqrt{5} & 8 - 2\sqrt{15} & 18\sqrt{3} - 14\sqrt{5} \\ 1 & -\sqrt{3} + \sqrt{5} & 8 - 2\sqrt{15} & -18\sqrt{3} + 14\sqrt{5} \end{bmatrix} = 1920 = 2^7 \times 3 \times 5$$

Let $\sqrt{3} + \sqrt{5} = \theta$. Therefore, we have to examine whether there exist α such that $\frac{1}{2}(\lambda_1 + \lambda_2\theta + \lambda_3\theta^2 + \lambda_4\theta^3)$ is an algebraic integer where $\lambda_k \in \{0, 1, 2, 3\}$. We first check the norm of it.

Exercise 2.5

i. Since the minimal polynomial is $x^4 - 2$, whose roots are $\sqrt[4]{2}, \omega\sqrt[4]{2}, \omega^2\sqrt[4]{2}, \omega^3\sqrt[4]{2}$ where ω is the 4th root of the unity. The 4 monomorphisms σ_k are those holding the value on $\mathbb{Q}$ and

$$\begin{cases} \sigma_1 : \sigma_1(\sqrt[4]{2}) = \sqrt[4]{2} \\ \sigma_2 : \sigma_2(\sqrt[4]{2}) = \omega\sqrt[4]{2} \\ \sigma_3 : \sigma_3(\sqrt[4]{2}) = \omega^2\sqrt[4]{2} \\ \sigma_4 : \sigma_4(\sqrt[4]{2}) = \omega^3\sqrt[4]{2} \end{cases}$$

ii. As we know from *i)*, the minimal polynomial is $x^4 - 2$.

iii. a) $\prod_{k=1}^4 (x - \sigma_k(\sqrt[4]{2})) = x^4 - 2$

b) $\prod_{k=1}^4 (x - \sigma_k(\sqrt{2})) = (x^2 - 2)^2$

c) $\prod_{k=1}^4 (x - \sigma_k(2)) = (x - 2)^4$

d) $\prod_{k=1}^4 (x - \sigma_k(\sqrt{2} + 1)) = (x^2 - 2x - 1)^2$

Exercise 2.6

When $p = 3$, we follow from the book by checking the trace first:

$$\mathrm{T}(\alpha) = \lambda_1 \in \mathbb{Z}, \text{ which is useless}$$

We then compute the norm:

$$\begin{aligned} \mathrm{N}(a + b\theta + c\theta^2) &= (a + b\theta + c\theta^2)(a + b\omega\theta + c\omega^2\theta^2)(a + b\omega^2\theta + c\omega\theta^2) \\ &= a^3 + 5b^3 + 25c^3 - 15abc \\ &\equiv 0 \pmod{27} \end{aligned}$$

Hence,

Exercise 2.7

Exercise 2.8

a) $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$

Exercise 2.9

Exercise 2.10

Suppose it is not an integral basis, and denote the additive group $\alpha_1, \ldots, \alpha_n$ generated by G. Therefore G must be a subgroup of $\mathfrak{O}$, and its discriminant cannot be equal to d.

Exercise 2.11

The problem is trivial since any monomorphisms in the problem holds the value in $\mathbb{Q}$.

Exercise 2.12

Let

$$\begin{cases} K_1 = \mathbb{Q}(x^6 - 1) \\ K_2 = \mathbb{Q}(x^3 - 1) \end{cases}$$

In this case, from *Exercise 11* we have

$$\begin{cases} \mathrm{N}_{K_1}(1) = 6 \\ \mathrm{N}_{K_2}(1) = 3 \end{cases}$$

Exercise 2.13

i. The n monomorphisms have the same forms and proofs as those in *Theorem 2.4*.

ii. $\mathrm{N}_{K/L}(\alpha_1\alpha_2) = \prod_{k=1}^n \sigma_k(\alpha_1\alpha_2) = \prod_{k=1}^n \sigma_k(\alpha_1) \prod_{k=1}^n \sigma_k(\alpha_2) = \mathrm{N}_{K/L}(\alpha_1)\mathrm{N}_{K/L}(\alpha_2)$
$\mathrm{T}_{K/L}(\alpha_1 + \alpha_2) = \sum_{k=1}^n \sigma_k(\alpha_1 + \alpha_2) = \sum_{k=1}^n \sigma_k(\alpha_1) + \sum_{k=1}^n \sigma_k(\alpha_2) = \mathrm{T}_{K/L}(\alpha_1) + \mathrm{T}_{K/L}(\alpha_2)$ $\square$

iii.

iv. *(I suppose the author means $\mathrm{N}_{K/L}(\alpha)$ since $\sqrt{\alpha}$ is not in K)*
Field K has minimal polynomial $x^2 - \sqrt{3}$, which has roots $\sqrt[4]{3}, -\sqrt[4]{3}$, over $\mathbb{Q}$.

   i. When $\alpha = \sqrt[4]{3}$, $\mathrm{N}_{K/L}(\alpha) = \sqrt[4]{3}(-\sqrt[4]{3}) = -\sqrt{3}$, $\mathrm{T}_{K/L}(\alpha) = \sqrt[4]{3} - \sqrt[4]{3} = 0$.

   ii. When $\alpha = \sqrt[4]{3} + \sqrt{3}$, $\mathrm{N}_{K/L}(\alpha) = (\sqrt[4]{3} + \sqrt{3})(-\sqrt[4]{3} + \sqrt{3}) = 3 - \sqrt{3}$, $\mathrm{T}_{K/L}(\alpha) = (\sqrt[4]{3} + \sqrt{3}) + (-\sqrt[4]{3} + \sqrt{3}) = 2\sqrt{3}$.

Exercise 2.14

i. $\mathrm{N}_{K/L}(\sqrt{3}) = \sqrt{3}\sqrt{3} = 3$, $\mathrm{N}_{K/\mathbb{Q}}(\alpha) = \sqrt{3}(-\sqrt{3})\sqrt{3}(-\sqrt{3}) = 9$.

ii. $\mathrm{T}_{K/L}(\sqrt{3}) = \sqrt{3} + \sqrt{3} = 2\sqrt{3}$, $\mathrm{T}_{K/\mathbb{Q}}(\alpha) = \sqrt{3} + (-\sqrt{3}) + \sqrt{3} + (-\sqrt{3}) = 0$.